

MIPS32 5-Stage Pipelined CPU

Project Overview

This project implements a 32-bit scalar processor based on Harvard architecture.

The CPU is compatible with the MIPS32 Release 1 instruction set architecture (ISA) and features a static 5-stage pipeline. The system systematically handles control hazards, data hazards, and structural hazards.

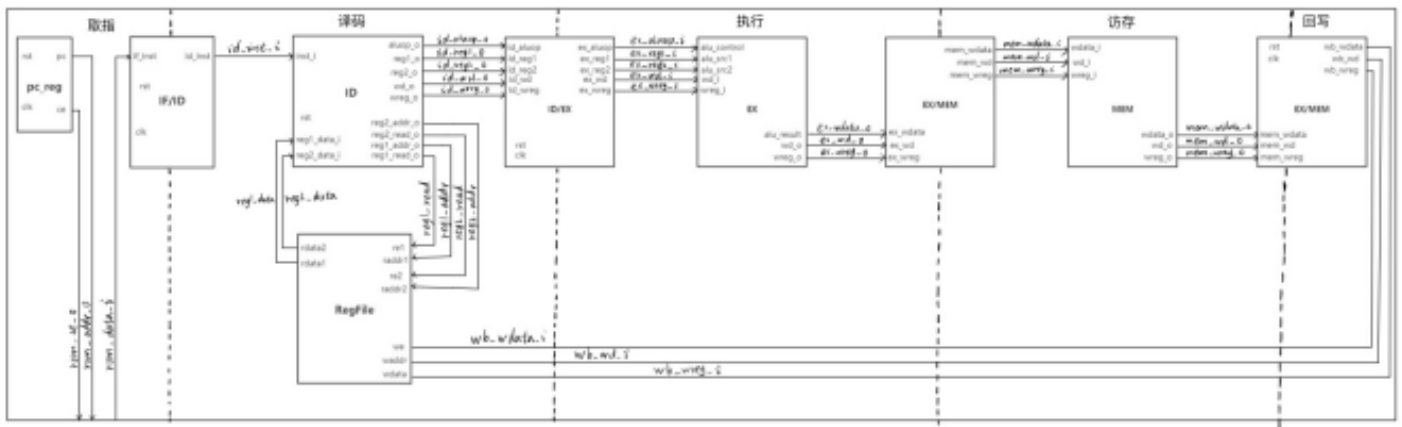
Development Environment & Tools

- **Hardware Environment:** Personal Computer (PC)
- **Software IDE:** Xilinx Vivado
- **Hardware Description Language (HDL):** Verilog

System Design

The pipeline is divided into five distinct operational stages, executing one specific part of an instruction per clock cycle:

- **IF (Instruction Fetch):** Retrieves the instruction from memory.
- **ID (Instruction Decode):** Translates the instruction and reads registers.
- **EX (Execute):** Performs arithmetic/logical operations.
- **MEM (Memory Access):** Reads from or writes to data memory.
- **WB (Write-Back):** Writes the computation result back to the register file.



Supported Instruction Set

The CPU successfully implements 21 core instructions, categorized into three standard MIPS formats:

- **Immediate Type (I-Type):** `ADDIU` , `LUI` , `BEQ` , `BNE` , `LW` , `SW`
- **Jump Type (J-Type):** `JAL` , `JR`
- **Register Type (R-Type):** `ADD` , `ADDU` , `SUB` , `SUBU` , `SLT` , `SLTU` , `AND` , `NOR` , `OR` , `XOR` , `SLL` , `SRA` , `SRL`

Pipeline Architecture

The top-level integration (`open_mips`) connects various sub-modules designed in Verilog to achieve seamless data flow across the pipeline:

- **Fetch & Decode Modules:** `pc_reg` provides the program counter logic, while `inst_rom` serves as the instruction memory. The `id` module decodes the fetched instructions, interacting with the `regfile` (General Purpose Registers) to acquire source operands.
- **Execution Modules:** The `ex` module functions as the core Arithmetic Logic Unit (ALU). It works alongside `hi_lo_reg` , which stores the results of specific multiplication or division operations.
- **Memory & Write-Back Modules:** The `mem` module governs interactions with the `data_ram` (Data Memory) during load/store operations.
- **Pipeline Isolation Registers:** Modules such as `if_id` , `id_ex` , `ex_mem` , and `mem_wb` act as buffers between stages, temporarily storing instruction addresses, operation types, and data to ensure isolation between clock cycles.
- **Global Control Unit:** The `ctrl` module is the central nervous system for pipeline management, controlling pipeline stalling and flushing actions based on execution requests.

Delay Slots Technology

To reduce the performance penalty caused by branch instructions, the processor implements Branch Delay Slot technology, allowing the instruction immediately following a branch to execute regardless of the branch outcome. Branch condition evaluation is also performed early in the Decode (ID) stage to further reduce pipeline flushing overhead.

Structural hazards and data dependencies are handled dynamically through pipeline stalling. The control module monitors pipeline states and temporarily halts upstream stages until data dependencies or structural conflicts are resolved.